

MamaCare

Maternal Health Risk Classification Capstone

Problem: classify maternal-health records as Low, Mid or High Risk.

Inputs: six routine measurements only.

Purpose: academic screening-support demonstration, not diagnosis.

Dataset: 1014 rows

Duplicates: 562

Conflicting signatures: 35

Data and Approach

Source: UCI Maternal Health Risk dataset.

Target: RiskLevel.

Leakage control: identical six-measurement signatures kept in the same split.

Feature engineering: PulsePressure, MeanArterialPressure, AgeBand.

Models compared: baseline, Logistic Regression, Random Forest, SVC, Gradient Boosting, XGBoost, MLP and Soft Voting.

Primary metric: Weighted F1.

Secondary metrics: Macro F1, High Risk recall, accuracy and log loss.

Results

Selected model: SoftVoting
Weighted F1: 0.508
Macro F1: 0.518
High Risk recall: 0.782
Baseline Weighted F1: 0.228

Class	Recall	F1
high risk	0.782	0.811
low risk	0.765	0.623
mid risk	0.090	0.119

Important weakness: Mid Risk recall remains low, so the model is not deployable for clinical decisions.

Limitations and Responsible Use

The dataset is small, duplicated and not locally validated in Kenya.

Some identical measurements have different labels.

Dataset input ranges are not clinical normal ranges.

The frontend avoids personal identifiers and warns against diagnosis or treatment use.

Fairness criterion: Equal Opportunity for High Risk identification.

Subgroup performance differs by age group.

Real use would require human review, monitoring and redress.

Conclusion and Next Steps

MamaCare is a complete classification capstone with frontend, backend, tests, documentation and responsible-use reporting.

The strongest technical choice is group-separated evaluation, which avoids inflated results from duplicates.

Next phase: collect locally validated maternal-health data with richer clinical context and evaluate the model prospectively before any real-world use.